HARIRAMAKRRISHNAN RAMACHANDRAN

+353 89 970 6156 • hariramakrrish@gmail.com • Dublin, Ireland • **Stamp 1G — Full-Time Work Eligible**

PROFESSIONAL EXPERIENCE — 3.5 YEARS

Software Engineer — HCL Technologies, Chennai, India

Sep 2021 – Jan 2025

- Designed and developed **machine learning models** using **PyTorch** on **GPU** infrastructure to detect anomalous payment behaviour, achieving an **AUC of 0.89** on a held-out dataset.
- Contributed to the full data science lifecycle, including data exploration, **feature engineering**, model development, and validation for financial crime prevention initiatives.
- Translated complex analytical findings into clear, actionable recommendations for **risk and compliance teams**, facilitating data-driven decision-making.
- Applied rigorous **statistical methods**, including hypothesis testing and regression analysis, to ensure model robustness and explainability for a regulated financial environment.
- Collaborated within a global, cross-functional team of data scientists and engineers across multiple regions to deliver on project objectives.
- Contributed to **engineering best practices**, including version control with Git, code quality standards, and ensuring reproducibility of experiments.
- Containerised and deployed **Python** data science applications to **AWS Lambda**, managing dependencies and ensuring efficient execution for data processing tasks.

SKILLS

Machine Learning – PyTorch, scikit-learn, Deep Learning, Supervised Learning, Unsupervised Learning, Model Evaluation, Model Validation

Programming Languages – Python, SQL, R

Data Science Libraries – pandas, NumPy, PyTorch, scikit-learn

Data Handling – Data Wrangling, Feature Engineering, Data Exploration, SQL, Data Warehousing

Cloud & Deployment – AWS (EC2, S3, Lambda, CloudWatch), Docker, GPU Computing, CUDA

Statistical Methods – Hypothesis Testing, Regression, Experimental Design, Probability

Collaboration & Communication – Cross-functional Teams, Technical Communication, Stakeholder Management

PROJECTS

Financial Crime Detection Model (MSc Data Analytics)

- Engineered a feature pipeline using **Python** and **SQL** to extract behavioral signals from a dataset of approximately 1.1 million transaction records.
- Trained a **gradient-boosted classifier** on the extracted features, tuning class weights to address a 1:140 fraud imbalance and achieving an **AUC of 0.91**.
- Developed and deployed the trained model behind a **FastAPI** inference endpoint, ensuring input validation and achieving a p95 latency of 180ms.

Real-time Transaction Anomaly Detection

- Designed and built **machine learning models** using **PyTorch** for detecting anomalous payment behaviour in high-volume transaction data.
- Leveraged **GPU/CUDA acceleration** to optimize model training and inference at scale, processing approximately 3,500 events per second during peak hours.
- Collaborated with **risk and product teams** to refine model performance and integrate anomaly alerts into existing operational workflows.

EDUCATION

M.Sc. Data Analytics — National College of Ireland, Dublin

Feb 2026

B.E. Computer Science — SNS College of Technology, Coimbatore, India 2021

CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)